Customer Churn Prediction using Machine Learning

1. Introduction
Customer churn prediction is a critical business problem where companies aim to identify customers likely to discontinue services. This project builds an end-to-end machine learning pipeline to predict customer churn using structured telecom data.

2. Dataset Description
The dataset was sourced from Kaggle (Telco Customer Churn Dataset). It contains customer demographic information, account details, subscribed services, and churn status. The target variable is Churn (Yes/No).

3. Data Cleaning & Preprocessing
Missing values were handled, irrelevant columns removed, and categorical features encoded using one-hot encoding. The target variable was converted into binary numerical format for modeling.

4. Exploratory Data Analysis (EDA)
EDA revealed that customers with month-to-month contracts and higher monthly charges were more likely to churn. Long-tenure customers showed strong retention behavior.

5. Model Development
Three machine learning models were trained and evaluated:
- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

6. Model Evaluation
Models were evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC curve. Random Forest achieved the highest overall performance and was selected as the final model.

7. Advanced Analysis
Feature importance analysis showed that contract type, tenure, and monthly charges were the most influential features. ROC-AUC score confirmed strong model discrimination capability.

8. Deployment
The best-performing model was saved as a pickle file and deployed using Streamlit, allowing users to interactively predict customer churn through a web interface.

9. Business Impact
By identifying high-risk customers, businesses can take preventive actions such as targeted offers and improved customer engagement, leading to reduced churn and increased revenue.

10. Conclusion
This project demonstrates a complete data science workflow from data preprocessing to deployment, showcasing practical application of machine learning in solving real-world business problems.